

FOKAM NINYIM ASTRIDE MELVIN

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Nationality: Cameroonian citizen **Languages:** French (Native), English, Deutsch

OBJECTIVE

I am seeking a research opportunity at the intersection of AI safety, global catastrophic risks, and sustainable development. I aim to apply my skills in machine learning, computer vision, and scientific computing to investigate high-impact problems in biosecurity, climate resilience, and robust AI systems.

PROFILE

Google DeepMind Scholarship - Data Scientist - Educator

- **Engaged AI researcher** with a passion for applying cutting-edge machine learning techniques to real-world challenges across climate forecasting, credit risk modeling, and biosecurity

Interests: Machine learning for science, AI for social impact, STEM for Africa, Sustainable Development Goals (SDGs)

EDUCATION

- **African Institute for Mathematical Sciences (AIMS)** January 2025 - November 2025
AI for Science Master's Programme Funded by Google DeepMind Cape Town, South Africa
- **University of Yaounde I** 2019-2024
Master degree in computer science (data science) Yaounde, Cameroon

HONORS AND AWARDS

- **Google DeepMind Scholarship** 2025
Google DeepMind
 - for the AI for Science programme
- **African Health Award** 2023
Digital Health Hackathon
 - Third Prize at the Hackathon with Intelligent Patient Administration System (IPAS)

RESEARCH EXPERIENCE AND PUBLICATIONS

- **Master Thesis – Robust Deep Learning for Climate Forecasting** ongoing
AIMS South Africa, Supervisor: Dr. Shruti Nath
 - Designed and tested ML architectures under non-stationary data conditions
 - Surveyed climate modeling approaches and uncertainty quantification techniques
 - Developed methods for multi-source and multi-scale data integration
 - Applied neural networks for complex time series analysis
- **Diagnosis of Plant Diseases Using Mobilenet and Inception Architectures. SSRN, 2024.**
 - <http://dx.doi.org/10.2139/ssrn.5021799>
 - Developed classification models to identify plant diseases from images
 - Applied transfer learning techniques to maximize performance with limited data

WORKSHOPS AND SCIENTIFIC OUTREACH

- **I-X AI in Science** April 2025 - May 2025
AIMS South Africa and Imperial College London [Notebooks](#) [🔊](#)
 - An interdisciplinary workshop exploring the application of artificial intelligence and machine learning to scientific research across fields such as chemistry, biology, physics, and mathematics.
 - Core Topics: Classification, clustering, deep learning, and domain-specific modeling techniques.
 - Hands-on Labs: Daily practical sessions using Python to implement and experiment with techniques introduced during lectures.
 - Applied Focus: Emphasis on how AI drives innovation in scientific discovery, combining theoretical insights with real-time coding exercises.

• Science and Engineering of Large Language Models (LLMs)

March 2025 - April 2025

AIMS South Africa ,Google DeepMind , MILA and OXFORD

Notebooks 

- Foundations of LLMs: From word2vec to transformers, tokenization techniques, and coding a dense transformer and attention mechanism from scratch.
- Training & Optimization: Pre-training strategies, efficient GPU training, custom training loops
- Advanced Architectures: Scaling laws, Mixture of Experts (MoE), and lab conversion from dense to MoE models.
- Specialized Topics: Diloco and asynchronous training techniques.
- Model Adaptation: LoRA (Low-Rank Adaptation), adaptors, and quantization (sparsity, weights, activations).
- Scalability: Profiling (roofline models, MATmuls), FSDP (Fully Sharded Data Parallel), and data parallelism.
- Post-Training: Instruction tuning and advanced post-training methods.
- Hands-on Labs: Emphasis on practical implementation and experimentation throughout the workshop.

• Modern Statistics and Machine Learning for Population Health in Africa

March 2025 - April 2025

AIMS South Africa and Imperial College London

Notebooks 

- Learned implementation of modern probabilistic programming languages (Stan, NumPyro) for statistical modeling
- Applied Bayesian methods to epidemiological, genomic, and spatial data analysis
- Developed models addressing regional challenges including disease outbreaks, air pollution, and extreme weather events
- Gained hands-on experience with automatic differentiation and gradient-based optimization techniques
- Implemented intelligence-driven statistical approaches for evidence-based public health applications
- Created analysis pipelines combining probabilistic programming with machine learning for healthcare applications
- Collaborated on air quality prediction using Stan and real PM2.5 data from Kampala, Uganda.

• ATLAS CERN International Masterclass — Particle Physics Outreach

March 2025

AIMS South Africa , IPPOG

- Guided high school students in identifying particles (electrons, muons, photons) using real data from CERN's ATLAS experiment. Supported team-based analysis and international video conference with CERN moderators.

PROJECTS

• Project A: Credit Risk Modeling with Machine Learning

Tools: Python, Scikit-learn, XGBoost, SHAP, Pandas, Matplotlib

- Built end-to-end machine learning models to predict loan default probabilities using real-world financial datasets.
- Performed feature engineering on borrower characteristics, credit history, and transactional features.
- Tuned XGBoost with cross-validation and GridSearchCV to optimize AUC-ROC performance (achieved 0.89).
- Used SHAP values for model explainability, identifying the most influential features for credit decisioning.
- Delivered actionable insights for financial institutions to improve lending strategies and mitigate risk.

• Project B: Climate Simulation and Downscaling with Neural Networks

Tools: Python, TensorFlow, NetCDF4, CMIP6 Data, Scikit-learn, Matplotlib

- Built neural network models to predict temperature and precipitation from CMIP6 global climate model outputs
- Applied data preprocessing using NetCDF files and spatial filtering for geospatial climate analysis
- Analyzed prediction bias and variance across multiple models
- Developed interpretable visualizations to inform adaptation strategies
- Implemented transfer learning techniques to improve model generalization

• Project C: Time Series Forecasting for Financial Market Volatility

Tools: Python, Facebook Prophet, LSTM (Keras), Scikit-learn, Pandas, Plotly

- Developed time series models to forecast short-term financial market volatility using historical SP 500 and macroeconomic indicators.
- Applied Facebook Prophet for trend and seasonality modeling, and LSTM networks for capturing long-term dependencies.
- Combined lag features and exogenous variables (macroeconomic indicators) for enriched predictive performance.
- Evaluated models using MAE, RMSE, and directional accuracy to support risk-sensitive investment decisions.
- Produced interactive dashboards for visualizing forecasts and uncertainty intervals for decision-makers.

• Project D: OCR project

Tools: Python, Tesseract OCR, AWS Lambda, Terraform, Docker, OpenCV

- Designed a cloud-based Optical Character Recognition (OCR) system to automate extraction of information from scanned documents
- Used Tesseract OCR with OpenCV for image preprocessing (binarization, noise removal, rotation correction).
- Deployed the pipeline using AWS Lambda for serverless processing and Terraform for infrastructure-as-code.
- Containerized the app with Docker for scalable and portable deployment
- Delivered a robust solution for digitization of administrative and financial records at scale.

SKILLS

- **Programming Languages:** Python, R, SQL,C, C++, javascript,java,ROS,Cuda
- **Web Technologies:** HTML, CSS, JavaScript, React, Flask,Django
- **Database Systems:** MySQL, PostgreSQL, MongoDB
- **Data Science & Machine Learning:** scikit-Learn, TensorFlow, Keras,Pytorch,Apache Spark,Jax,SHAP,LIME,OpenCV
- **Cloud Technologies:** AWS, Google Cloud Platform, Microsoft Azure
- **DevOps & Version Control:** Git, Docker, Jenkins, Kubernetes, Ansible,Streamlit
- **Transferable Skills for Bioinformatics:** Large-scale and heterogeneous data processing, Integration of multiple data sources, Development of predictive models with uncertainty quantification, Implementation of reproducible analysis pipelines,Mathematical Modeling,Exploration of models to generate testable hypotheses
- **Mathematical & Statistical Tools:** MATLAB,Oracle, Excel, Statistical Analysis System (SAS), SPSS, Mathematica
- **Other Tools & Technologies:** Tableau, Power BI, Jupyter Notebook, Apache Spark, Hadoop,HYPATIA,Prophet
- **Research Skills:** Data Collection and Analysis, Statistical Modeling, Literature Review, Report Writing, Presentation

MEMBERSHIPS

- **Young AI Leaders Community, AI for Good**
- **Association International des Etudiants en Sciences Economique et Commercial(AIESEC)**
- **Applied Machine Learning Day's(AMLD) Africa**
- **Zindi | AfriClimate AI | Black In AI | Deep Learning Indaba**

REFERENCES

- **Dr. Claire David**, AI for Science Master's Academic Director, African Institute for Mathematical Sciences. claired@aims.ac.za
- **Dr. Shruti Nath** , Supervisor, Oxford University. shruti.nath@physics.ox.ac.uk